

Financial Question and Answer Bot

By:

FinOclik (Team A1)



Name	Student Number	Email	Role
Janit Jindal	500211232	Janitjindal@loyalistcollege.com	Team Lead/ NLP advisor And NLP/Backend developer
Kiranpreet Kaur	500211233	Kiranpreetkaur11@loyalistcollege.com	Front end developer/UI advisor
Megha Ann Mathew	500211185	Meghaannmathew@loyalistcollege.com	NLP/Backend developer

Loyalist College in Toronto

Date of completion : April 19,2024

Signature of Team Members :

Kiranpreet Kaur : *Kiranpreet Kaur*

Janit Jindal : *Janit Jindal*

Megha Ann Mathew : *Megha Ann Mathew*

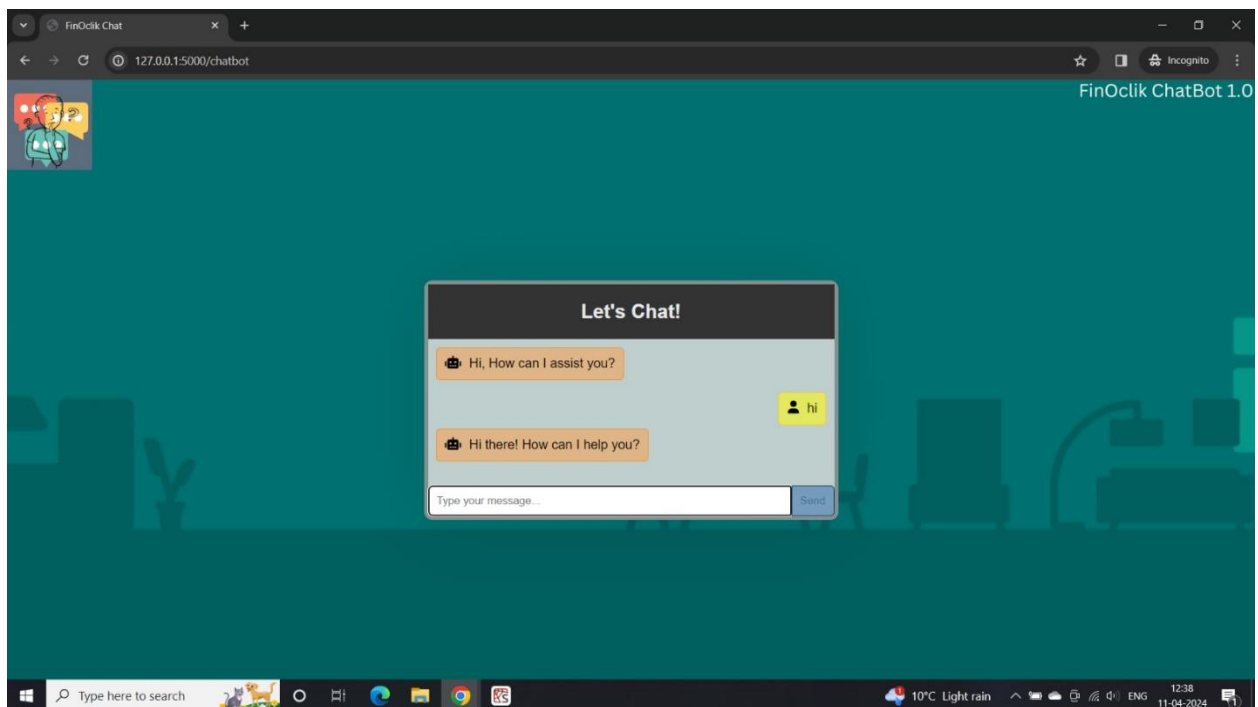
Abstract

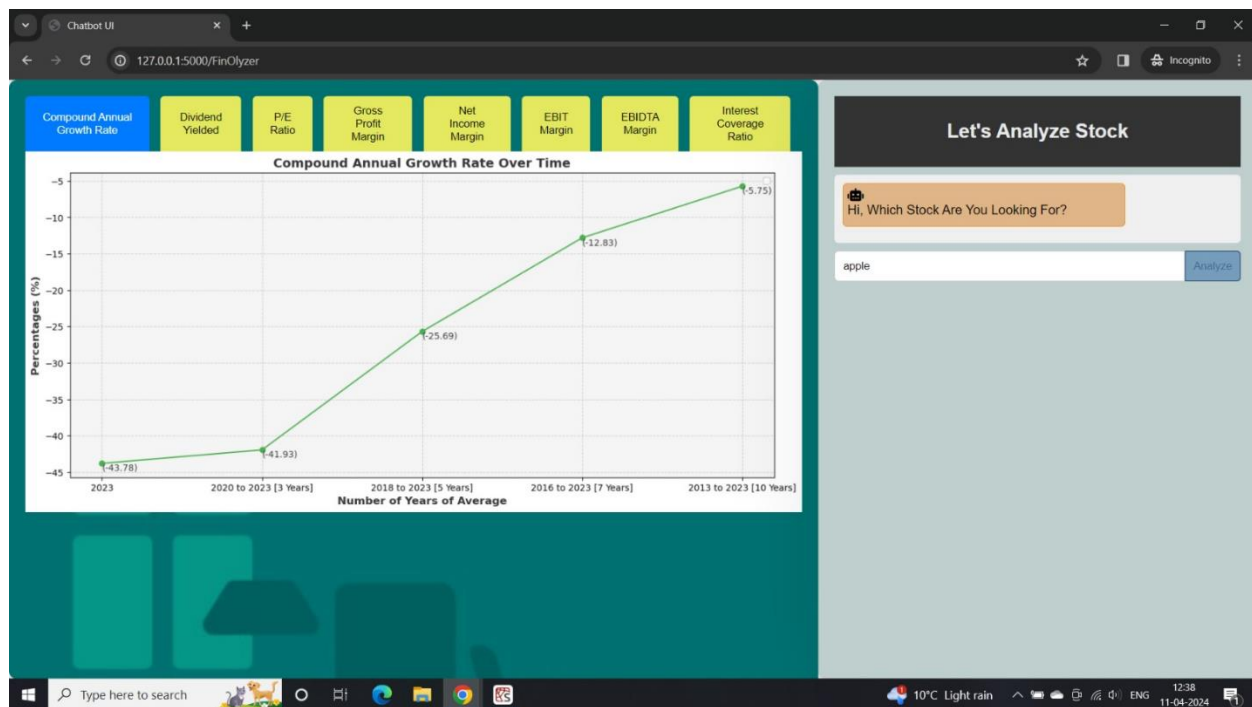
- **Summary of the project:**

We FinOclik has worked upon making a chatbot which can help the customers to enhance their knowledge in Finance. After gaining this, they can use their analytical thinking to make the decisions for financial related problems in their lives.

However, our chatbot help you in queries that can be related to the investment tips, getting knowledge of the companies where money can be invested and more about the stocks such as the current situation of apple and many more.

Our main goal is to provide impactful product where our customers can explore the world of financial independence and train minds to unlock analytical skills to make personalized investment decisions anytime and anywhere with a single click.





- **Problem Statement:**

Individuals in today's turbulent economy do not have access to comprehensive and simplified financial education tools, which leads to confusion and anxiety about money management and wealth accumulation. The fragmented nature of financial information on the internet exacerbates the issue, making it difficult for clients to build a comprehensive knowledge of fundamental concepts and methods. As a result, many people fail to make informed financial decisions and effectively build their portfolio of assets, which impedes their journey to financial independence. There is an urgent need for a centralized platform that collects and simplifies financial knowledge, giving users with the required assistance and resources to navigate the intricacies of the financial landscape and achieve their long-term financial objectives.

- **How did we solve it with technology?**

General Finance Questions:

FinOclik offers a specific part or service where users can ask general finance questions and receive expert guidance and recommendations.

The software provides personalized solutions based on the user's individual condition and financial goals, allowing them to make informed decisions.

Stock analysis:

FinOclik is a comprehensive stock analysis tool or service that enables customers to evaluate and analyse a variety of stocks and investment options.

Table of contents

1. Introduction
 - 1.1. Problem Statement
 - 1.2. Background information needed for the user
 - 1.3. Technologies Used
2. Literature Review
 - 2.1. Competing applications or technologies
3. Methods
 - 3.1. Technologies used (In Detail)
 - 3.2. Methods used to gather these type of data
 - 3.3. Reasons for using these types of method
4. Findings
 - 4.1. Findings from surveying audience.
 - 4.2. Findings from our research and technology advancement.
5. Discussion
 - 5.1. All features of our product.
 - 5.2. How our research impacts your technology.
 - 5.3. How our findings impact our technology.
 - 5.4. How your surveys impact your technology.
6. Conclusion
7. Recommendations
8. References

Introduction

- **Problem statement:**

People frequently find it difficult to get access to thorough and user-friendly financial education resources in the current economic climate. The lack of easily accessible resources causes uncertainty and worry over money management and wealth accumulation. The fragmented online financial information further exacerbates the problem by making it difficult for people to have a strong understanding of financial concepts.

Because of this, a lot of people struggle to effectively diversify their investment portfolios or make wise financial decisions. People's journey towards financial independence is hampered by the absence of easily available financial education since they might not be able to handle the complexity of the financial world without professional assistance.

In addition, a lack of trustworthy sources feeds false information and misconceptions about investing and personal finance. Without reliable information, people can become victims of fraud or make poor judgements that could harm their financial situation.

All things considered, the situation of financial education today is far from ideal, with people having difficulty finding trustworthy sources and figuring out the intricacies of personal finance. In order to enable people to take charge of their financial destinies and work towards reaching their long-term financial goals, it is imperative that these issues be resolved.

- **Background information that is needed for the reader:**

Following is the background knowledge a user has to be aware of before accessing the chatbot:

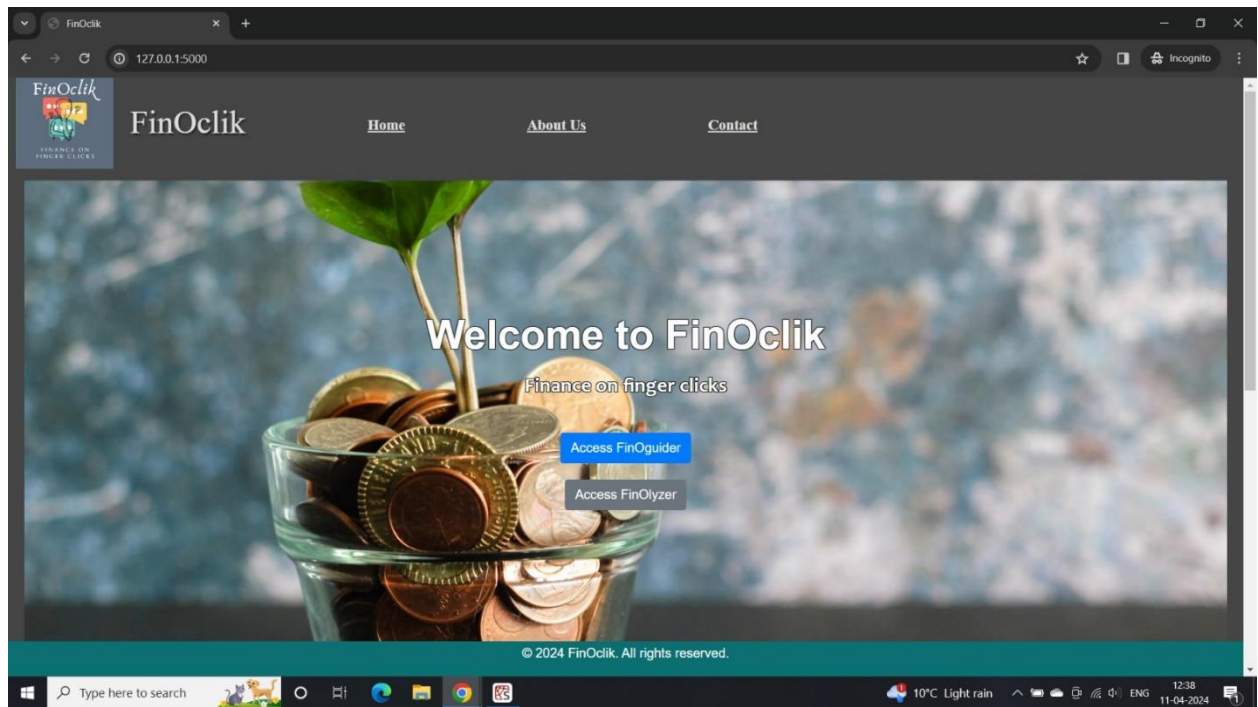
- **Functionality:** The chatbot that answers financial questions is meant to help consumers with questions about general financial knowledge, investment techniques, personal money, and budgeting advice.
- **Expertise:** Although the chatbot has extensive financial expertise, it's crucial to remember that it is an AI-powered instrument rather than a certified financial advisor. Users who are making complex financial decisions should use critical thinking skills and consult a professional.

- **Data security and user privacy** are given top priority by the chatbot's programming. Beyond the duration of the current session, no personal information is stored, and all communications are encrypted to maintain confidentiality.
- **Educational Resource:** The chatbot not only responds to specific questions but also acts as an instructional tool by providing clarifications and insights into a range of financial ideas in order to increase users' financial literacy.

Limitations: Although the chatbot makes an effort to deliver pertinent and correct information, it might not always have the background knowledge necessary to answer extremely specialised or niche financial queries. In certain situations, users might need to seek advice from experts or other resources.

- **Technologies Used:**

- Natural Language Processing (NLP)
- Machine Learning (ML)
- Artificial Intelligence (AI)
- Chatbot Frameworks (Flask)
- Programming Languages (Python)
- Web APIs (Application Programming Interfaces)
- Web Development Technologies (HTML, CSS)
- User Interface Design Tool (WIX)



Have a look in our financial website.

Literature Review

- **Competing applications and technologies:**

Erica from Bank of America:

As the virtual assistant for Bank of America, Erica helps clients with their regular banking needs. In order to complete operations like paying bills, managing transactions, transferring money between accounts, and checking balances, users can communicate with Erica. By providing rapid and easy access to financial services via a conversational interface that can be accessed through the Bank of America website or mobile app, it seeks to simplify the banking experience.

Kasisto:

Kasisto provides chatbots driven by AI that are designed specifically for financial institutions to deliver virtual assistant services to their clients. These chatbots can help users with a variety of tasks, including product recommendations, money management across several accounts, account reviews, and payment processing. The ultimate goal of Kasisto's chatbots is to improve customer experience by providing individualized help and support.

PayPal:

PayPal users can get customer service support via its chatbot by using Facebook Messenger. It helps with things like maintaining account holds and limitations, resolving refunds, changing passwords, and monitoring the status of payments.

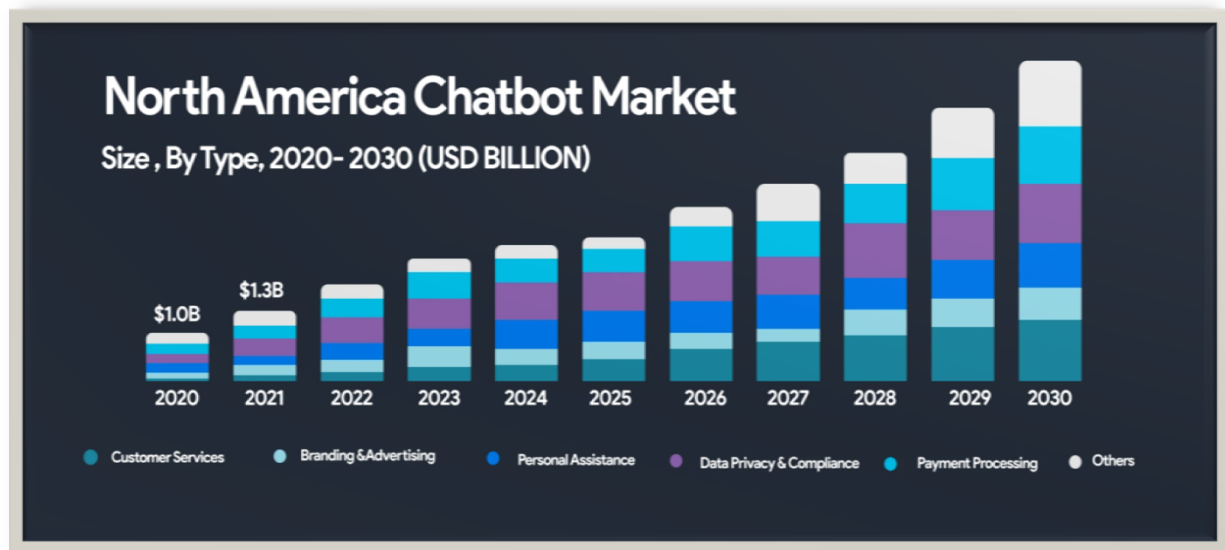
PayPal seeks to improve the user experience by providing prompt and effective solutions to users' questions and concerns by means of a chatbot interface for customer care.

AI Passage:

Financial companies can develop conversational chatbots with Passage AI to help consumers with a range of financial chores and inquiries. These chatbots can help with credit card enquiries, loan application support, account information access, and updates on financial goods and services.

(Stefanowicz, 2024)

(Top Chatbots in Financial Services for 2022 | Botpress Blog, n.d.)



Prasannan, A. (2023, November 22). *The Rise of Chatbot Banking: Understanding its Potential*. Mindster-Blogs. <https://mindster.com/mindster-blogs/the-rise-of-chatbot-banking-understanding-its-potential>

Methods

- **Technologies Used:**

NLP, or natural language processing

The goal of natural language processing (NLP), a subfield of artificial intelligence, is to empower machines to comprehend, translate, and produce human language. It includes methods for processing and examining speech or text data in natural language in order to find patterns, derive meaning, and improve human-machine communication. NLP is utilized in many different contexts, such as text summarization, sentiment analysis, language translation, and chatbots.

Machine Learning (ML)

Building statistical models and algorithms that allow computers to learn from data and make predictions or judgements without explicit programming is known as machine learning (ML), a subset of artificial intelligence. Large datasets are used to train machine learning (ML) algorithms, which subsequently find patterns and relationships that can be utilized to automate processes or generate predictions.

Intelligent artificial systems (AI)

In computer science, artificial intelligence (AI) is a broad field that focuses on developing intelligent computers that are capable of doing activities that normally require human intelligence. Artificial Intelligence (AI) comprises a range of approaches and techniques, such as robotics, computer vision, natural language processing, machine learning, and expert systems. Applications of AI include financial forecasts, medical diagnosis, driverless cars, and virtual assistants.

Flask chatbot frameworks

Software development tools or platforms that offer libraries, APIs, and utilities for creating, implementing, and maintaining chatbots are known as chatbot frameworks. Flask is a lightweight Python web framework that reduces overhead for creating web applications, including chatbots. It may be used to create conversational interfaces and integrate with other NLP systems because it has features like routing, request management, and template rendering.

Python Programming Languages

Python is a popular programming language that is well-known for being easy to learn, versatile, and simple. For a wide range of applications, including as web development, data analysis, machine learning, and artificial intelligence, it offers large libraries and frameworks. Python is a popular choice for creating chatbots, NLP apps, and other AI-related projects because of its large ecosystem and ease of use.

Application Programming Interfaces, or Web APIs

Web APIs are interfaces that provide online communication and interaction between various software systems. Through requests and responses in a standard format (JSON or XML), they let developers access and utilize the features of distant services or applications. In web development, web APIs are frequently used to integrate third-party services, obtain data from external sources, and build interactive web applications.

Technologies for Web Development (HTML)

Web developers employ HTML (Hypertext Markup Language) as foundational technologies for building and designing web pages. The foundation of web development is laid by HTML which are also necessary to create contemporary, responsive, and aesthetically pleasing websites and web apps.

WIX, the User Interface Design Tool

WIX is a well-known cloud-based platform that lets users customize templates and utilize drag-and-drop tools to construct websites. Without the need for coding expertise, it offers an intuitive interface for creating and modifying web pages. With the help of WIX, you can create visually appealing and responsive websites with a choice of design components, multimedia integration, e-commerce capabilities, and mobile optimization.

- **Methods used to gather any type of data (surveys, research, etc.) and Reasons for using these methods.**

APIs

Utilized financial APIs provided by reputable sources such as Alpha Vantage to gather real-time financial data including stock prices, market trends, historical data, and company financials. As these APIs often offer endpoints for accessing a wide range of financial data in a structured format.

Reason for using the APIs was as they offer direct access to structured, real-time financial data. They also provided us a practical means of programmatically retrieving data, such stock prices, market movements, and corporate financials, making it easy to integrate them into our chatbot.

Web Scraping

Extracted data from financial websites, news portals, and regulatory filings using web scraping techniques. With web scraping, we have retrieved information from news a variety of financial websites such as Investopedia, Harvard Business School, RBC Royal Bank Website and many more. It offered versatility in obtaining particular data points or information that APIs do not deliver.

Surveys and Questionnaires

Utilized surveys from the students, our project advisor and program coordinator of the college to learn more about the financial requirements, preferences, and their habits. Surveys were created mostly verbally by asking them individually about the same. Even we have improved the functionality and user experience of our chatbot by gaining vital insights from the analysis of the responses.

However, during our expo we got to know more interests of the customers and feedback from them to get better features for our product.

These Surveys facilitated us with the understanding of users' requirements, preferences, and spending patterns. Then we customized our chatbot's features and responses based on direct user feedback which helped us to better meet the demands of our customers and boost user happiness.

Access industry studies, whitepapers, and research reports that have been released by regulatory agencies, financial institutions, and consulting firms. These documents frequently included insightful information about financial analysis techniques, investment strategies, and market patterns.

Social Media Monitoring

Keep an eye out for conversations, opinions, and trends pertaining to money and investments on social media sites like LinkedIn and Twitter.

Machine Learning Models

Utilizing past financial data, train models to forecast stock prices, recognize market patterns, or categorize the tone of financial news. Libraries such as TensorFlow and scikit-learn can be used to create and implement predictive models.

Findings from Surveying Audiences

- **Investment Preferences:** According to the poll, our readership has a wide range of preferences for investments, from cautious to risky. Certain users place more importance on stability and long-term growth than others, while others are more drawn to situations with high risk and big profit.
- A considerable segment of our readership has a proclivity towards stock research and market expertise. To make well-informed investing selections, they look for knowledge on spotting prospective stocks, comprehending market trends, and evaluating company performance.
- **Need for Financial Advice:** Our audience has a clear need for financial advice and help, as indicated by the study. A significant number of users indicate a wish to obtain individualized advice and suggestions based on their unique financial objectives and risk appetite.
- **Want for Educational Content:** Users say they would rather have access to content that improves their knowledge of investing and finance principles. They appreciate materials like articles, tutorials, and guidelines that help them become more financially literate and make more informed investing decisions.
- **Importance of User Experience:** It becomes clear that consumers' engagement and happiness with the platform are significantly influenced by their user experience. The significance of user-friendly navigation, lucid communication, and prompt assistance is emphasized by users during their chatbot engagements.

Discussion

Our Product and its Features

- **Investment Advice:** Depending on the user's time horizon, investment goals, and risk tolerance, the chatbot can provide tailored investment advice. Based on our customer preferences and market conditions, it can recommend a range of investment vehicles, including stocks, bonds, mutual funds, exchange-traded funds (ETFs), and more.
- **Stock Suggestions:** Users can ask questions concerning particular stocks or industries and receive suggestions on which stocks are gaining ground or have the potential to gain a lot of value. To make wise recommendations, the chatbot can examine market trends, historical data, and business fundamentals.
- **Financial inquiries:** Users are able to ask common financial inquiries, like how to manage debt, set up a budget, or save money for retirement. The chatbot can offer information and tailored guidance to assist users in making wiser financial decisions.
- **Stocks Analysis:** Using a visual representation of NASDAQ company data is a really good concept. Users can view comparisons, trends, and stock performance via interactive charts and graphs. The chatbot can provide valuable insights about key metrics such as revenue growth, earnings per share, and price-to-earnings ratio.
- **Educational Content:** In addition to responding to queries, the chatbot can provide educational material on a variety of financial topics.
- **Performance Analytics:** With the use of visual performance analytics tools, investors may monitor and assess the evolution of their assets over time. Features like graphical representations and analysis of past performance could be included to evaluate investment returns and help customers make wise selections.

How our research impacts your technology

- **Algorithm Development:** The ability of the chatbot to comprehend and appropriately react to user inquiries is directly influenced by research into machine learning models and natural language processing (NLP) algorithms. We can continuously improve the chatbot's language

comprehension capabilities, enhancing its conversational flow and user experience, by keeping up with the most recent developments in NLP research.

- **Methods of Data Analysis:** Our ability to evaluate financial data and give users with insights is influenced by research in machine learning and data analytics. Our ability to recognize patterns, trends, and correlations in financial markets allows us to apply state-of-the-art data analytic techniques to provide consumers with more knowledgeable stock recommendations and investing advice from the chatbot.
- **Sentiment Analysis:** Studies on sentiment analysis and natural language comprehension enable us to ascertain the opinions of the general public and the market regarding particular stocks, businesses, or market patterns. We can evaluate investors' general mood by integrating sentiment analysis into the chatbot's functionality, and by doing so, we may modify our recommendations to better assist users in making well-informed investment decisions.
- **Security & Privacy:** Cybersecurity and data privacy research guides the development of security protocols that safeguard users' personal and financial information. By keeping up with the most recent security risks and best practices, we can make sure that the chatbot platform complies with legal and industry regulations, protecting users' private information from abuse or unauthorized access.
- **Market Trends and Industry Insights:** Studies on financial markets, economic indicators, and industry trends offer insightful background information that guides the chatbot's suggestions and reactions. We can give consumers timely, pertinent information and insights to help them traverse the challenges of financial decision-making by keeping an eye on market trends and being up to date on industry advancements.

How your findings impact our technology

- **Enhanced Natural Language Understanding:** We can improve our chatbot's natural language processing (NLP) skills thanks to research insights. We continuously enhance the chatbot's comprehension of user queries with accurate and contextual understanding by examining linguistic patterns and nuances, as well as by analyzing user interactions and

comments. As a result, users receive more insightful responses and have a more seamless conversational experience.

- **Better Recommendation Algorithms:** Our study results are taken into consideration when creating the recommendation algorithms that the chatbot uses to offer stock analysis and investing advice. We improve our algorithms to give people recommendations that are more relevant and tailored to them by looking at market trends, historical data, and user preferences. This aids consumers in making well-informed financial decisions catered to their own requirements and objectives.
- **User Experience (UX) Optimization:** Research results help us make our chatbot platform's UX (user experience) even better. The layout, navigation, and functionality of the chatbot can be improved by identifying areas of user behavior, preferences, and pain points through research studies and usability testing. We are able to improve the user experience iteratively in this way, giving people a more pleasant, intuitive, and engaging experience.
- **Research Results:** Our chatbot platform's features and content are informed by insightful data that is obtained through research. Our analysis of market trends, user interactions, and performance indicators provides us with important insights into the needs, preferences, and behaviors of our users. Due to this, we are able to better serve our target audience and provide more value by customizing the chatbot's services, recommendations, and content.
- Our development team is able to maintain a culture of constant innovation because of our research-driven methodology. We always work to improve the capabilities of our chatbot by keeping up with the most recent developments in financial markets, technology, and user behavior. In order to maintain our chatbot's position at the forefront of innovation in the financial services sector, we must investigate new features, integrate cutting-edge technologies, and adjust to shifting market conditions.

How your surveys impact your technology

- **Recognizing User wants:** Surveys offer important information about user preferences, wants, and problems with financial services and investments. Through survey input, we are

able to better understand the unique demands and challenges of our consumers and customize the features and functionality of the chatbot to better suit their needs.

- **Collecting Input on Current Features:** Through surveys, we can obtain input regarding the current features of the chatbot platform and pinpoint opportunities for enhancement. We can prioritize improvements and upgrades that will have the biggest effects on user experience and happiness by asking users about their satisfaction levels, usability problems, and feature requests.
- **Verifying Assumptions:** User behavior and preference-related assumptions and hypotheses are verified with the use of surveys. We may verify whether our design choices meet user expectations and modify our strategy by putting our presumptions to the test with survey questions. This guarantees that the functionalities and features of the chatbot are based on actual user requirements and preferences.
- **Determining Market Trends:** Through surveys, we may gather information on more general market trends and advancements in the sector. Through inquiring about customers' investment objectives, approaches, and perspectives on financial products, we are able to detect new patterns and changes in consumer perception that could impact our technology roadmap and product strategy.
- **Users' demographic and psychographic information** is gathered through surveys, which helps with user segmentation and personalization. We can further customize the chatbot's recommendations and content to meet the needs and tastes of various user categories by segmenting consumers based on variables like age, income, risk tolerance, and investment goals.
- **User happiness Measurement:** Surveys are a useful technique for gauging users' loyalty to the chatbot platform and their level of happiness. We can monitor changes over time and pinpoint areas where modifications are required to improve overall user satisfaction and retention by tracking user satisfaction levels and their propensity to suggest the platform to others through regular surveys.
- **Iterative Improvement:** Our chatbot technology is improved continuously with the help of surveys. Through iterative feedback collection and integration into our development cycle, we can gradually enhance the chatbot's functionality, user experience, and performance. This

approach guarantees that the chatbot adapts to users' evolving requirements and expectations.

Conclusion

Discussions above that need resolution

- **Data collection:** Use the chatbot's UI to include survey prompts that ask users about their experiences using the platform. For instance, ask customers to score their happiness and add any more remarks or recommendations after offering advice on investments or responding to a financial question.
- **Data analysis:** To find trends in sentiment, reoccurring topics, and opportunities for development, compile survey responses and analyse them methodically. Pay attention to important data including feedback sentiment, user satisfaction scores, and frequently mentioned problems as reported by users.
- Sort survey responses according to their importance to the goals of the chatbot, their effect on user satisfaction, and their practicality for deployment. Give top priority to fixing usability problems that impede user interaction or improving features that get constant good feedback from users.
- Transform survey results into recommendations and actionable insights for enhancing the user experience and operation of the chatbot. Depending on the input, this can entail improving conversational scripts, introducing fresh investment or learning tools, or resolving particular user issues.
- **Implementation:** Working with the development team, apply modifications and improvements that are informed by the comments from the survey. Assemble a roadmap for the chatbot's evolution that incorporates feedback-driven enhancements, and make sure that upgrades are included in later platform versions.
- **Observation and Iteration:** Keep an eye on how newly introduced modifications are affecting user feedback, engagement metrics, and satisfaction. Refine analysis methods, gather more survey data, and make new enhancements based on continuous user feedback to iterate on the feedback loop.

To sum up, our financial chatbot platform's continuous improvement process requires the incorporation of survey results in order to increase user experience, functionality, and efficacy. We may systematically gather, evaluate, and act upon user feedback to promote ongoing improvement

and optimization of the chatbot's features and capabilities by putting in place an organized feedback loop.

We can priorities improvements that solve the most urgent issues and provide the most value to users by using surveys to gather insightful data about user needs, preferences, and pain areas. By converting survey results into useful information, we can decide on feature creation, usability enhancements, and content optimization.

Working together with cross-functional teams guarantees that the development plan for the chatbot incorporates and efficiently executes feedback-driven enhancements. We can measure how changes we apply affect user happiness, engagement metrics, and platform performance overall through continuous monitoring and iteration, which helps us make future improvements and refinements more effectively.

Overall, including survey feedback enables us to develop a financial chatbot platform that is user-focused, attentive to user requirements, and always adapting to satisfy investors' shifting expectations. We want to provide a smooth and worthwhile user experience that gives users the confidence to make wise financial decisions by using this iterative development method.

Conclusion of the Project

The goal of our project is to enhance users' understanding of stocks and finance by creating a sophisticated financial chatbot platform. Our chatbot offers a wide range of services to serve as a virtual financial advisor, providing customers with solutions to a variety of financial questions. Our platform gives customers personalized suggestions based on their risk tolerance and financial goals, from asking for investment tips to finding stocks with strong returns. To help consumers make well-informed decisions, we also place a high priority on providing them with in-depth stock analysis and NASDAQ company data presented in an eye-catching manner. Our chatbot fosters financial literacy by delivering precise, timely, and actionable insights through the smooth integration of modern natural language processing and data analytics. This helps users manage the complexities of the financial system.

To provide impactful product where our customers can explore the world of finance and train minds to unlock analytical skills to make personalized investment decisions anytime and anywhere with a single click.

References:

Top chatbots in financial services for 2022 | Botpress Blog. (n.d.). <https://botpress.com/blog/top-chatbots-financial-services>

S. (2024, January 3). *How AI-Driven Chatbots are Transforming the Financial Services Industry*. <https://www.linkedin.com/pulse/how-ai-driven-chatbots-transforming-financial-services-industry-ugrmc/>

Drozдов, A. (2023, September 1). *Chatbots in Finance: Benefits, Examples, and Future*. Yellow. <https://yellow.systems/blog/chatbots-in-finance>

Financial Terminology: 20 Financial Terms to Know. (2018, October 11). Business Insights Blog. https://online.hbs.edu/blog/post/finance-for-non-finance-professionals-finance-terms-to-know?c1=GAW_CM_NW&source=CA_GEN_PMAX&cr2=content-ca-gen-pmax&kw=general&cr5=&cr6=&cr7=c&utm_campaign=content-ca-gen-pmax&utm_term=general&gad_source=1&gclid=Cj0KCQjwq86wBhDiARIsAJhuphmuInQMO98mQGjGBISGLvzQkfs3EUUOkYQkHus1cJriuP4Lzgd8DNUaAhObEALw_wcB

O'Shea, A., & Voigt, K. (2024, March 22). *Stock Market Basics: What Beginner Investors Should Know*. NerdWallet. <https://www.nerdwallet.com/article/investing/stock-market-basics-everything-beginner-investors-know>

Five AI Technologies. (n.d.). SAS. <https://www.sas.com/en-ca/insights/articles/analytics/five-ai-technologies.html>

Financial Terminology: 20 Financial Terms to Know. (2018, October 11). Business Insights Blog. <https://online.hbs.edu/blog/post/finance-for-non-finance-professionals-finance-terms-to-know>

Financial Terms A-Z. (2019, September 10). Investopedia. <https://www.investopedia.com/financial-term-dictionary-4769738>

Glossary of Terms for Newcomers to Canada. (2023, May 9). RBC Royal Bank. <https://www.rbcroyalbank.com/en-ca/new-to-canada/glossary-financial-terms/>

N. (n.d.). *Nasdaq: Stock Market, Data Updates, Reports & News*. Nasdaq. <https://www.nasdaq.com/>

Yahoo is part of the Yahoo family of brands. (n.d.). <https://finance.yahoo.com/quote/%5EIXIC/>

Bloomberg - Are you a robot? (n.d.). <https://www.bloomberg.com/quote/CCMP:IND>

Financial Times. (n.d.). Financial Times. <https://www.ft.com/>

Strickland, K. (n.d.). *Customer Service AI Chatbots - Freshchat by Freshworks.*

https://www.freshworks.com/customer-service-suite/lp/ai-conversational-chatbots/?tactic_id=5587714&utm_source=GoogleAdwords&utm_medium=Fchat-Search-US-East-Generic-BOFU-Chatbot&utm_campaign=Fchat-Search-US-East-Generic-BOFU-Chatbot&utm_term=chatbot%20in%20software&device=c&matchtype=p&network=g&gclid=Cj0KCQjwlN6wBhCcARIsAKZvD5gKElQPgkMG0dcf9xuE73Mn9C1fwECFUuf31ahqqGG4cyuQfj850kaAo3BEALw_wcB&audience=kwd-1249032152042&ad_id=640030498218&gad_source=1

Prasannan, A. (2023, November 22). *The Rise of Chatbot Banking: Understanding its Potential.* Mindster-Blogs. <https://mindster.com/mindster-blogs/the-rise-of-chatbot-banking-understanding-its-potential>

Wikimedia Foundation. (2024, April 9). Compound annual growth rate. Wikipedia. https://en.wikipedia.org/wiki/Compound_annual_growth_rate

Financial terminology: 20 financial terms to know. Business Insights Blog. (2018, October 11). <https://online.hbs.edu/blog/post/finance-for-non-finance-professionals-finance-terms-to-know>

Bowers, M. (2020, January 22). A guide to chatbot terminology (with infographic): Toptal®. Toptal Design Blog. <https://www.toptal.com/designers/chatbot/chatbot-terminology>

Tarver, E. (n.d.). How is a company's share price determined with the Gordon Growth Model? Investopedia. <https://www.investopedia.com/ask/answers/061615/how-companys-share-price-determined.asp>

Here's how inflation affects stocks: 5 things you need to know. Here's how inflation affects stocks: 5 things you need to know | Voya.com. (n.d.). <https://www.voya.com/blog/heres-how-inflation-affects-stocks-5-things-you-need-to-know>